

Leakage-Resistant Rice Leaf Disease Classification Under Class Imbalance: An Empirical Study of Moderated Class Weighting and Cross-Split Generalization

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Abstract

Deep learning has shown promise for rice leaf disease classification, but reported performance can be difficult to compare when datasets, class distributions, acquisition conditions, and evaluation procedures differ. This study examines the effect of class weighting under a leakage-resistant evaluation protocol using the RiceLeafDiseaseBD Version 2 dataset. The original 9,769 images were audited with exact-content hashing. The audit identified 39 duplicate groups involving 78 files, including eight groups in which identical image content had conflicting labels. After removing conflicting-label copies and redundant same-class duplicates, 9,722 unique images remained. These images were split stratifiably into 7,777 training, 972 validation, and 973 locked held-out test images. A frozen ImageNet-pretrained MobileNetV2 classifier was evaluated with three training strategies: no class weighting, full inverse-frequency weighting, and moderated weighting obtained by taking the square root of the full weights. Macro-F1 was used as the primary model-selection criterion, and The test set was not considered when picking the models. Modulated weighting did best on the validation dataset delivering an accuracy of 64.61 percent a Macro-F1 of 58.41 percent and a Weighted-F1 of 63.22 percent. In the test dataset that was excluded from the model selection the chosen model reached an accuracy of 60.12 percent a Macro-F1 of 53.14 percent and a Weighted-F1 of 58.24 percent. The main issue is the low F1-score, for leaf smut, which is 11.54 percent while Sheath Blight, Rice Tungro and Healthy scored 74.52 percent, 71.25 percent and 69.32 percent respectively. The validation-to-test Macro-F1 decline of 5.27 percentage points shows that validation performance alone can give an optimistic picture of generalization. The findings suggest that the strength of class weighting affects the balance between minority-class recognition and overall performance, while dataset integrity and class-wise held-out evaluation remain essential for interpreting agricultural AI models.

Keywords: Rice disease classification; MobileNetV2; class imbalance; class weighting; data leakage; duplicate auditing; generalization; agricultural AI; computer vision

1. Introduction

Rice is a major food crop, and disease outbreaks can reduce yield and increase the cost and effort required for crop management. Deep learning and computer vision are increasingly used to recognize disease symptoms from leaf images. In field settings, however, images can vary in illumination, background, leaf orientation, disease severity, and image quality. These factors make it important to evaluate models under conditions that are closer to real agricultural imagery rather than relying only on a single overall accuracy value.

A second concern is data integrity. If the same image, or an exact copy of it, appears in more than one partition, the reported test result can be overly optimistic. The problem is especially relevant to agricultural image collections because several photographs may be taken from the same plant or scene. Previous work on image-classification benchmarks has shown that duplicate contamination can affect estimates of generalization [2].

This study takes a methodological approach to these issues. Instead of introducing a new network architecture, it uses MobileNetV2 as a lightweight and reproducible backbone and concentrates on three questions: whether exact duplicate auditing changes the usable dataset, how the strength of class weighting affects class-wise and

aggregate performance, and how the selected model behaves on a locked held-out test set. MobileNetV2 was chosen because its inverted residual blocks and linear bottlenecks were designed for efficient visual recognition [3].

The experiments use RiceLeafDiseaseBD Version 2, a field-image dataset collected in Gazipur, Bangladesh, containing six classes: Blast, Brown Spot, Healthy, Leaf Smut, Rice Tungro, and Sheath Blight [4]. The dataset is not a Nepalese field dataset. Nepal is used only as the broader application context, so the present results should not be interpreted as evidence of performance on Nepalese rice fields.

2. Related Work

2.1 Rice disease image classification

Paddy Doctor contains 16,225 field images across 13 classes and reported a ResNet34 F1-score of 97.50% under its benchmark protocol [5]. More recent studies have investigated lightweight models and attention mechanisms. Xu et al. reported 95.68% test accuracy for LiSA-MobileNetV2 on a different Paddy Doctor-derived experimental setup [6]. These results demonstrate the potential of deep learning for rice disease recognition, but direct numerical comparison is difficult because datasets, class definitions, acquisition conditions, and evaluation protocols differ.

2.2 Dataset reliability and generalization

Recent reviews of AI-based rice disease detection emphasize dataset provenance, class imbalance, evaluation design, and deployment readiness [1,7]. Newer datasets have also been collected under field conditions, including BanglaRiceLeaf, which contains 4,152 images across five categories [8]. The continued growth of such datasets makes it increasingly important to distinguish improvements caused by model design from improvements caused by easier data or evaluation settings.

2.3 Class imbalance

Class imbalance can make a classifier favor well-represented categories while giving insufficient attention to minority classes. Class weighting changes the loss contribution of each class so that errors on underrepresented categories have greater influence during training. Full inverse-frequency weighting can be useful, but it can also move the decision boundary too strongly toward minority classes. A moderated weighting scheme therefore provides a simple way to examine whether a smaller correction gives a better balance.

3. Research Gap and Objectives

Many rice disease studies report high classification scores, but the combination of exact-duplicate auditing, controlled weighting comparisons, and locked within-dataset testing is less consistently reported. This study addresses that methodological gap rather than claiming a new state-of-the-art architecture.

The objectives were to:

- (1) audit the dataset for exact duplicates and conflicting labels.
- (2) create a clean and reproducible train-validation-test split.
- (3) compare unweighted, full inverse-frequency, and moderated class weighting with the same MobileNetV2 architecture.
- (4) select the model using validation Macro-F1 without using the test set.
- (5) examine held-out performance, uncertainty, and class-specific errors.

Research questions:

RQ1: How does exact-duplicate and conflicting-label auditing affect dataset reliability?

RQ2: How does class weighting affect MobileNetV2 performance on the six-class dataset?

RQ3: Can moderated class weighting provide a better balance between minority-class recognition and overall performance than full inverse-frequency weighting?

RQ4: How well does the selected model generalize to a locked held-out test set, and which classes show the largest weaknesses?

4. Materials and Methods

4.1 Dataset

RiceLeafDiseaseBD Version 2 contains 9,769 high-resolution 1024×1024 images collected with smartphones in rice fields in Gazipur, Bangladesh, during the Summer 2025 growing season. It covers six classes and includes natural field backgrounds and acquisition conditions [4]. The present study used the original-image classification folders rather than the dataset's object-detection annotations.

Class	Original	Clean	Clean (%)
Blast	1,326	1,323	13.61%
Brown Spot	2,178	2,176	22.38%
Healthy	1,575	1,573	16.18%
Leaf Smut	724	721	7.42%
Rice Tungro	2,244	2,238	23.02%
Sheath Blight	1,722	1,691	17.39%
Total	9,769	9,722	100%

Table 1. Original and cleaned class distribution.

All 9,769 images were processed with exact-content hashing. This produced 9,730 unique image contents, 39 duplicate hash groups, and 78 files involved in duplicate groups. Eight groups contained identical image content under different class labels. Because an exact duplicate cannot provide two independent visual observations, both copies in a conflicting group were removed rather than choosing a preferred label by visual inspection. Same-class duplicate groups were reduced to one representative.

In total, 47 files were removed: 16 from conflicting-label groups and 31 redundant same-class copies. The resulting dataset contained 9,722 images. A second audit found no remaining exact duplicate groups and no remaining conflicting-label groups. Exact hashing does not, however, rule out near-duplicates or correlations caused by multiple images from the same plant or acquisition session [2].

5. Results

5.1 Validation Comparison

Model	Accuracy	Macro-F1	Weighted-F1
Baseline	62.96%	55.38%	61.01%
Full Class Weighting	60.49%	56.87%	60.41%
Moderated Class Weighting	64.61%	58.41%	63.22%

Table 4. Validation performance of the three training strategies.

Moderated class weighting gave the best validation results among the three strategies, with 64.61% accuracy, 58.41% Macro-F1, and 63.22% Weighted-F1. Compared with the unweighted baseline, the gains were 1.65,

3.03, and 2.21 percentage points, respectively. Full weighting increased Macro-F1 by 1.49 percentage points but reduced accuracy by 2.47 percentage points and Weighted-F1 by 0.61 percentage points.

5.2 Per-Class Validation Performance

Class	Baseline F1	Full weighting F1	Moderated F1
Blast	40.72%	44.08%	44.36%
Brown Spot	57.36%	44.95%	59.54%
Healthy	67.60%	69.01%	72.83%
Leaf Smut	8.43%	27.65%	22.22%
Rice Tungro	72.85%	71.30%	72.49%
Sheath Blight	72.02%	74.76%	79.01%

Table 5. Validation F1-score by disease class.

Full weighting raised Leaf Smut F1 from 8.43% to 27.65%, but Brown Spot F1 fell from 57.36% to 44.95%. Moderated weighting recovered Brown Spot performance and gave the highest F1 of the three strategies for Blast, Brown Spot, Healthy, and Sheath Blight. The results therefore show a trade-off rather than a uniform benefit from stronger weighting.

5.3 Experiment 3 Training Dynamics

The selected Experiment 3 run continued for all 12 epochs. Training and validation loss decreased over the run, while validation accuracy increased from 46.81% at epoch 1 to 64.61% at epoch 12. Validation accuracy being higher than training accuracy is expected here because the training pipeline used random augmentation and class-weighted loss, while validation images were evaluated without random augmentation and under the standard evaluation objective.

5.4 Held-Out Test Performance

Metric	Held-out test
Accuracy	60.12%
Macro Precision	53.04%
Macro Recall	54.75%
Macro-F1	53.14%
Weighted-F1	58.24%

Table 6. Final held-out-test performance of the selected model.

Class	Precision	Recall	F1
Blast	43.61%	43.94%	43.77%
Brown Spot	56.79%	42.20%	48.42%
Healthy	62.89%	77.22%	69.32%
Leaf Smut	18.75%	8.33%	11.54%
Rice Tungro	66.80%	76.34%	71.25%
Sheath Blight	69.39%	80.47%	74.52%

Table 7. Held-out-test performance by class.

The selected model was evaluated once on the locked 973-image held-out test set. Accuracy fell from 64.61% on validation to 60.12% on test, and Macro-F1 fell from 58.41% to 53.14%. Sheath Blight had the highest test F1 at 74.52%, followed by Rice Tungro at 71.25% and Healthy at 69.32%. Leaf Smut remained the weakest class, with an F1-score of 11.54%; six of its 72 test images were correctly classified.

5.5 Validation-to-Test Generalization

Class	Validation F1	Test F1	Drop
Blast	44.36%	43.77%	-0.59 pp
Brown Spot	59.54%	48.42%	-11.12 pp
Healthy	72.83%	69.32%	-3.51 pp
Leaf Smut	22.22%	11.54%	-10.68 pp
Rice Tungro	72.49%	71.25%	-1.24 pp
Sheath Blight	79.01%	74.52%	-4.49 pp

Table 8. Validation-to-test F1 generalization.

The performance drop was not uniform across classes. Brown Spot and Leaf Smut had the largest F1 declines, while Blast and Rice Tungro were comparatively stable. Overall Macro-F1 decreased by 5.27 percentage points. This difference shows why validation performance alone should not be treated as the final estimate of unseen-sample performance.

5.6 Error Analysis

The most frequent test-set confusion was Brown Spot → Rice Tungro (43 images), followed by Brown Spot → Blast (39). Other frequent errors included Blast → Rice Tungro (25), Rice Tungro → Brown Spot (24), Sheath Blight → Healthy (21), Brown Spot → Healthy (20), Blast → Brown Spot (19), and Leaf Smut → Healthy (19). In the normalized confusion matrix, only 8.3% of Leaf Smut images were correctly classified; 26.4% were predicted as Healthy and 22.2% each as Blast and Brown Spot.

5.7 Bootstrap Confidence Intervals

Metric	Estimate	95% Bootstrap CI
Accuracy	60.12%	57.14–63.10%
Macro-F1	53.14%	50.17–56.05%
Weighted-F1	58.24%	55.02–61.46%

Table 9. Bootstrap confidence intervals for final test metrics (5,000 resamples, seed 42).

The final test estimates were 60.12% accuracy (95% CI: 57.14–63.10%), 53.14% Macro-F1 (95% CI: 50.17–56.05%), and 58.24% Weighted-F1 (95% CI: 55.02–61.46%). In the paired validation bootstrap, moderated weighting improved Macro-F1 over baseline by 3.03 percentage points, with a 95% bootstrap interval of +0.22 to +5.80 percentage points. The moderated-versus-full-weighting difference was +1.54 percentage points, with a 95% interval of -1.15 to +4.16 percentage points, which includes zero.

6. Discussion

6.1 Effect of Weighting Strength

The three experiments show that the magnitude of class weighting changes the balance of errors. Full inverse-frequency weighting helped minority-class recognition, particularly Leaf Smut, but it also reduced performance for Brown Spot and lowered aggregate accuracy. Moderated weighting produced the best validation balance in this experiment. This does not establish square-root weighting as a generally optimal rule; it was the best of the three strategies evaluated with this dataset and architecture.

6.2 Leaf Smut Remains Difficult

Leaf Smut remained difficult even after weighting. The cleaned training set contained 577 Leaf Smut images, compared with 1,790 Rice Tungro and 1,741 Brown Spot images. A descriptive class-count analysis gave a Spearman correlation of 0.543 between training count and validation F1, with $p = 0.266$ across six classes. This is not statistically significant and cannot establish a causal relationship between class size and performance. Visual variability, symptom similarity, disease severity, illumination, background, and annotation quality may also contribute.

6.3 Brown Spot and Cross-Class Confusion

Brown Spot showed a large validation-to-test decline and was frequently confused with Rice Tungro and Blast. These errors may reflect visual similarity or distribution differences, but the current dataset and experiments do not contain the controls needed to determine the cause. The interpretation is therefore descriptive rather than causal.

6.4 Comparison with Published Results

The final test scores are lower than several headline results reported in rice disease classification studies. Paddy Doctor reported a 97.50% F1-score for ResNet34 under its own benchmark protocol [5], and LiSA-

MobileNetV2 reported 95.68% test accuracy on a different setup [6]. These numbers should not be treated as direct baselines because the datasets, class definitions, image sources, preprocessing, architectures, and split procedures differ. The contribution of the present study is instead its emphasis on data integrity, weighting trade-offs, and held-out generalization.

6.5 Implications for Agricultural AI

The model was substantially more reliable for Healthy, Rice Tungro, and Sheath Blight than for Leaf Smut. Such uneven performance matters for decision-support applications because an aggregate score can hide important failure modes. Future deployment studies should therefore include class-wise sensitivity, confidence calibration, and, where appropriate, an abstention option rather than forcing a prediction for every image.

7. Limitations

The study has several limitations. First the dataset uses a dataset collected in Bangladesh so the dataset does not establish performance on Nepalese rice fields. Second the held-out test set is independent of the training and validation subsets but comes from the underlying dataset; the dataset therefore measures within-dataset generalization rather, than cross-dataset or cross-country generalization. Third, only MobileNetV2 and three weighting strategies were evaluated. Fourth, exact hashing removes pixel-identical duplicates but does not guarantee that near-duplicates or correlated images from the same acquisition event have been removed. Finally, the Leaf Smut test subset contains only 72 images, which limits the precision of its class-specific estimate.

8. Future Work

Future work should include external validation with independently collected field images, ideally across Nepalese rice varieties, locations, and seasons. Group-aware or session-aware splitting could reduce correlations among images from the same plant or scene. Perceptual hashing or embedding-based similarity analysis could complement exact hashing. Other imbalance methods, including focal or balanced-focal loss and targeted augmentation, could be compared with moderated weighting. Calibration, uncertainty estimation, explainability, and abstention should also be evaluated before field deployment.

9. Conclusion

This study evaluated rice leaf disease classification under class imbalance using a leakage-resistant data preparation and evaluation protocol. Exact-content auditing reduced the original 9,769-image dataset to 9,722 unique images and removed all identified exact duplicate and conflicting-label groups. Three controlled MobileNetV2 training strategies were then compared. Moderated class weighting gave the strongest validation performance, reaching 64.61% accuracy, 58.41% Macro-F1, and 63.22% Weighted-F1. On the locked held-out test set, the selected model achieved 60.12% accuracy and 53.14% Macro-F1, with a 5.27-percentage-point validation-to-test Macro-F1 decline.

The main result is not that moderated weighting solves rice disease classification. Instead, the experiments show that the strength of imbalance correction changes class-level behavior and that a locked held-out evaluation reveals weaknesses that validation results can hide. Leaf Smut and Brown Spot were the most challenging classes, while Sheath Blight, Rice Tungro, and Healthy were more reliably recognized. Further work should focus on external field validation and uncertainty-aware evaluation before such models are considered for dependable agricultural decision support.

10. Data and Reproducibility Statement

The study used RiceLeafDiseaseBD Version 2. The dataset version and DOI are given in Reference [4]. The public research repository and archived Zenodo release provide the manuscript, figures, result summaries, citation metadata, and reproducibility documentation. The original dataset and trained model checkpoints are not redistributed. The reported experiments use the cleaned 9,722-image dataset and the final 7,777/972/973 split, with the held-out test set locked before final evaluation.

11. References

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